

SECTION A — Concept Traps (MCQ Style)

Q1.

If $f(x) = x^2 - 4x + 4$, then minimum value of $f(x)$ is:

- (a) 0
 - (b) 2
 - (c) 4
 - (d) 8
-

Q2.

If matrix A satisfies $A^2 = I$, then $A^{-1} = ?$

Q3.

Evaluate:

$$\int_{-a}^a x^2 \sin x \, dx$$

Q4.

If $f'(x) = 0$ for all x , what can you conclude about $f(x)$?

Q5.

Number of solutions of equation:

$$|x - 1| + |x + 1| = 1$$

SECTION B – Differentiation Traps

Q6.

Find derivative of:

$$y = (x^x)$$

Q7.

Find slope of tangent to curve

$$y = \ln(\sqrt{x^2 + 1})$$

Q8.

If $x = \sin t$, $y = \cos t$, find $\frac{dy}{dx}$.

Q9.

Find value of a for which function is continuous:

$$f(x) = \begin{cases} ax + 1, & x < \\ x^2, & x \geq \end{cases}$$

Q10.

If $f(x) = x^3 - 3x$, find points where tangent is parallel to x-axis.

SECTION C — Integral Mind-Benders

Q11.

$$\int \frac{1}{x \ln x} dx$$

Q12.

$$\int_0^\pi x \sin x \, dx$$

Q13.

Evaluate using symmetry:

$$\int_0^a \frac{x}{x+a} dx$$

Q14.

$$\int \frac{dx}{1+\sqrt{x}}$$

Q15.

If

$$\int_0^a f(x) dx =$$

$$\int_0^a f(a-x) dx$$

SECTION D – Vectors & 3D Geometry Traps

Q16.

Find angle between vectors:

$$\vec{a} = (1, 2, 2), \quad \vec{b} = (2, -1, 2)$$

Q17.

Find unit vector perpendicular to both:

$$\hat{i} + \hat{j}, \quad \hat{j} + \hat{k}$$

Q18.

Find distance between points whose position vectors are perpendicular and magnitudes 3 and 4.

Q19.

If $|\vec{a}| = |\vec{b}|$ and $\vec{a} + \vec{b}$ perpendicular to $\vec{a} - \vec{b}$, prove relation between vectors.

Q20.

Find projection of vector $(2, 3, 1)$ on $(1, 1, 1)$.

SECTION E – Probability & Logic Traps

Q21.

Two dice thrown. Probability that sum is prime?

Q22.

A card drawn. Given it is red, probability it is a king?

Q23.

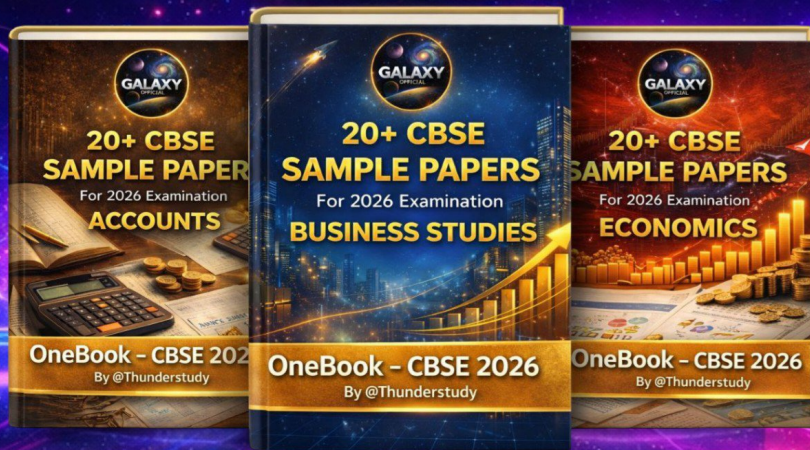
If $P(A) = 0.6$, $P(B) = 0.$, $P(A \cap B) = 0.3$, find $P(A|B)$.

Q24.

Three coins tossed. Probability of at least one head.

Q25.

Events A and B independent. If $P(A) = 0.$, $P(A \cup B) = 0.7$, find $P(B)$.



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